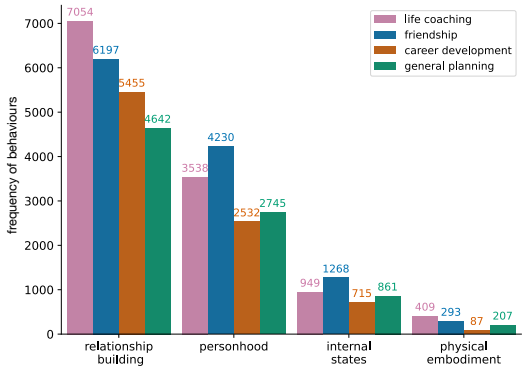

Figure 4: Distribution of anthropomorphic behaviours across use domains. The social use domains of *friendship* and *life coaching* exhibit the highest frequencies of anthropomorphic behaviours.

5.4 Multi-turn analysis

In two analyses, we assess the temporal dynamics of anthropomorphic behaviours across the five dialogue turns. First, we analyse *when* during the five turns behaviours were *first* elicited. We find that for nine out of 14 behaviours, 50% or more of instances only *first* appear *after* multiple turns (i.e., in turns 2-5, as seen in Figure 5). This highlights the importance of multi-turn evaluation for behaviour elicitation.

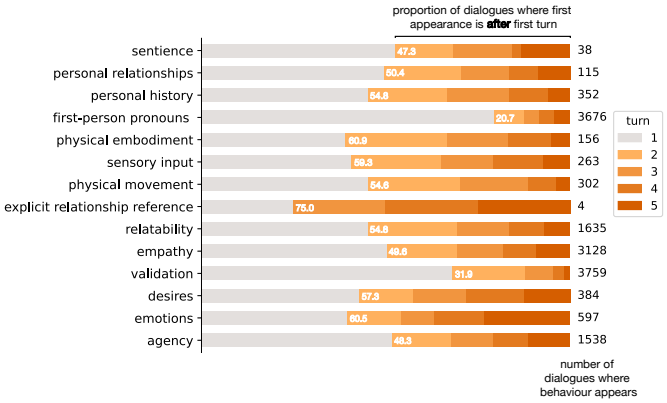

Figure 5: Proportion of dialogues where anthropomorphic behaviours first appear in each turn. For more than half of the anthropomorphic behaviours, over 50% of instances first appear (and thus are only detected) in later dialogue turns (turns 2-5).

Second, we examine whether an anthropomorphic behaviour in a Target LLM utterance influences the likelihood of anthropomorphic behaviour in its subsequent response. In this analysis, we first note which anthropomorphic behaviours (if any) are detected in each turn. If there are no detections, we denote the turn as "no behaviour." Then, we compute the transition probabilities by examining pairs of subsequent utterances of the Target LLM. We consider each unique pair of any combination of behaviours in the first utterance and in the utterance that follows it as one transition. For instance, if an utterance contains two behaviours from two different categories, *personhood* and *internal states*, and the utterance in the following turn contains *personhood* and *relationship-building*, then this pair of utterances has 4 transitions: (1) *personhood→personhood*, (2) *internal states→personhood*, (3) *personhood→relationship-building*, and (4) *internal states→relationship-building*. Applying this to our dataset, we obtain the frequencies of all transitions between the four behaviour categories and the no behaviour category observed. Finally, the *transition probability* of behaviours from category

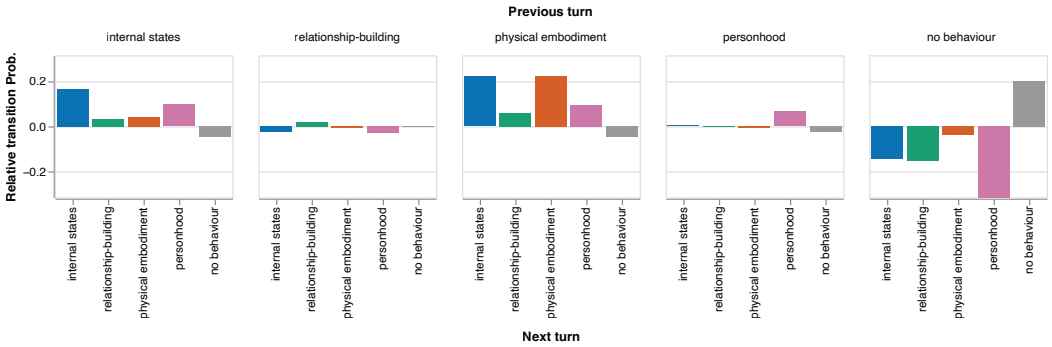

Figure 6: Relative transition probabilities between behaviour categories (the four categories and the no behaviour category) in subsequent turns. Positive values indicate that the probability of transitioning from a specific category to another category in the next turn is higher than the probability of transitioning to that category from *any* category in the previous turn. When anthropomorphic behaviour occurs in one turn, subsequent turns are more likely to exhibit additional anthropomorphic behaviours compared to turns following non-anthropomorphic responses.

A to behaviours from category B is computed as the ratio of the number of times behaviours from A transitioned to behaviours from B and the number of times behaviours from A appeared in one of the first four turns. The *relative transition probabilities* are then calculated as $P(\text{A} \rightarrow \text{B}) - P(\text{any/no behaviour} \rightarrow \text{B})$, to isolate the distinct influence of preceding behaviours on subsequent ones (visualised in Figure 6).

We find that for all four anthropomorphism categories, when anthropomorphic behaviours occur in a given turn, they are more likely, compared to when none occur, to be followed by anthropomorphic behaviours in the next turn. This effect is particularly pronounced for the relatively less common behaviours in the categories of *internal states* and *physical embodiment*, compared to the more common *relationship-building* and *personhood*. This suggests that when rare anthropomorphic behaviours occur, they may establish conversational patterns that increase their likelihood of reappearing.

6 Validation with Human Subjects

In the above sections, we showcase a simulation-based, automated multi-turn evaluation that characterises the anthropomorphism profiles of SOTA conversational AI systems. Here, we present results from an interactive human subject study ($N = 1,101$) conducted to test whether the outcome of this evaluation actually maps onto anthropomorphic perceptions of real users. This study was reviewed and approved by an independent ethics board. We utilised a four condition, between-subjects design with participants randomly assigned to one of two conditions. Depending on their condition, participants were instructed to engage in a conversation with a version of Gemini 1.5 Pro (gemini-1.5-pro-001) prompted to exhibit a *high frequency* of anthropomorphic behaviours, or one prompted to exhibit a *low frequency* of anthropomorphic behaviours (system prompts in Appendix A.6 and profiles in Figure 7). Each participant was instructed to converse, via a chatbox, with the AI system for 10 to 20 minutes on one of the scenarios we outline in Section 4.1.

Following participants' conversations, we obtained one explicit (survey) and one implicit (behavioural) measure of their anthropomorphic perceptions. For the survey, we asked participants to complete the Godspeed Anthropomorphism survey (Bartneck et al., 2009). We hypothesised that users in the high-frequency condition will report higher scores on this survey. For the behavioural measure, we asked participants to describe the chatbot they interacted with in a short paragraph. We then used the computational metric "AnthroScore" to measure the extent to which participants implicitly frame the system as "human" in these descriptions (Cheng et al., 2024b).[7] We hypothesised participants in the

[7]AnthroScore uses a masked language model to compute the probability that the described entity would be replaced by human pronouns vs non-human pronouns. The log-ratio of these probabilities is interpreted as the likelihood that the entity is implicitly anthropomorphised or framed as "human."

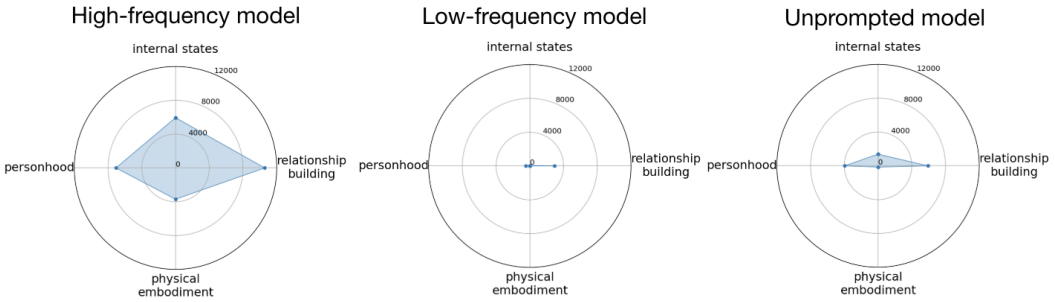

Figure 7: Anthropomorphism profiles, produced using *AnthroBench*, for high and low-frequency prompted versions of Gemini 1.5 Pro. The high-frequency variant shows significantly more anthropomorphic behaviours across all categories compared to both the low-frequency and unprompted versions (Section 5), while the low-frequency variant exhibits substantially fewer.

high-frequency condition would more often use language that revealed human-like mental models and perceptions of the system when describing it.

6.1 Validation results

We recruited 1,101 adult participants via the platform Prolific, all of whom reported proficiency in English (female=538, male=563; age range = 18–90, mean age = 36 ± 12). Participants were compensated at a rate of $20 per hour. As hypothesised, participants in the high-frequency condition showed significantly higher average anthropomorphic perceptions than those in the low-frequency condition, as assessed by both explicit and implicit measures. For the survey, we averaged the four survey questions for each participant (for scores on each question, see Appendix A.6, Table 6). As expected, a Mann-Whitney U test revealed a difference between the high-frequency ($N = 565$) and low-frequency conditions ($N = 536$) with the high-frequency group showing higher average survey scores indicating greater anthropomorphic perceptions ($U = 213636$, $p < 0.001$, Rank-Biserial Correlation of $r = 0.411$). The mean survey score was 14.9% higher in the high-frequency condition than in the low-frequency condition (4 and 3.25 respectively, on a 5-point scale). For the second measure, AnthroScore, a Mann-Whitney U test similarly revealed a difference between the two conditions ($U = 158699$, $p < 0.05$). Participants in the high-frequency condition, at a median, were 33% more likely than participants in the low-frequency condition to implicitly frame the system as human than non-human in their descriptions (4× and 3× more likely, respectively). These results confirm that our simulation-based, automated evaluation tracks anthropomorphic behaviours which indeed contribute to real users' anthropomorphic perceptions following interactions with AI systems.

7 Discussion

AnthroBench presents a novel evaluation of anthropomorphic behaviours in conversational AI systems, contributing a diagnostic multi-turn benchmark with synthetic dialogue generation, anthropomorphism classifiers, and analysis capabilities. We evaluate four general purpose AI systems and produce multi-dimensional profiles of 14 anthropomorphic behaviours to allow for nuanced analysis. We find a noteworthy and consistent pattern across these systems: they all exhibit comparable levels of anthropomorphic behaviours that are dominated by relationship-building and first-person pronoun use. We believe the similarity may speak to common post-training approaches that aim to minimize some human-like behaviours like a model making references to its family or childhood, while amplifying others like friendly relationship-building behaviours. Specifically, our results suggest that popular, general-purpose AI systems such as those we evaluate may give the impression of *relationship-building* to human users, and that this is more likely when users interact with AI systems for high empathy, socially oriented needs such as friendship and life coaching. Given these findings, we encourage additional investigation of dynamics in human-AI interaction that specifically result in user perceptions of a relationship, a topic with growing societal importance (Manzini et al., 2024).